



MID TERM EVEN SEMESTER EXAMINATION-2026

Roll no.

Name of the Program & Semester: B.Tech II SEM.

Name of the Course: Programming for Problem solving

Course Code: TCS201

Time: 1.5 hours

Maximum Marks: 50

Note:

(i) Answer all the questions except Que(2) by choosing any one of the sub-questions.

(ii) All 5 sub-questions in Que(2) are compulsory.

(iii) Each question carries 10 marks.

Que(1)	(10 marks)															
(a)	Explain compile time and run time initialization of 2-D Arrays with an example for each using a C code. Draw a flowchart to read two binary-2-D arrays A and B of size M-rows and N-columns, where the elements are only 1s and 0s. Find & display the resultant 2-D array C by bitwise ANDing the elements of the A and B arrays.															
OR																
(b)	Develop a code to input a square matrix of order $m \times n$. Find sum of each row of matrix having odd index.	CO1														
<table><tr><td>Sample Input:</td><td>Sample Output:</td></tr><tr><td>Row Size: 4</td><td>Sum of row index 1: 7</td></tr><tr><td>Column Size: 3</td><td>Sum of row index 3: 15</td></tr><tr><td>Inputted Matrix: 2 4 1</td><td></td></tr><tr><td>6 2 9</td><td></td></tr><tr><td>3 7 5</td><td></td></tr><tr><td>8 3 2</td><td></td></tr></table>			Sample Input:	Sample Output:	Row Size: 4	Sum of row index 1: 7	Column Size: 3	Sum of row index 3: 15	Inputted Matrix: 2 4 1		6 2 9		3 7 5		8 3 2	
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Que(2)	Predict the output of the following programs.	(10 marks)														
<table><tr><td>1. #include <stdio.h> void main() { int i, m, n, s=0; int mat[3][3]={1,2,3,4,5,6,7,8,9}; for(i=0;i<3;i++) s=s+mat[i][i]; printf("The sum is %d", s); }</td><td>2. #include <stdio.h> int main() { int a[] = {10, 20, 30, 40}; int *p = &a[3]; int *q = &a[1]; printf("%ld", p - q); return 0; }</td></tr><tr><td>3. #include <stdio.h> int main() { int a[] = {1, 2, 3}; int *p = a; printf("%d %d", ++*p, *p); return 0; }</td><td>4. #include <stdio.h> int main() { char str[] = "abcdef"; printf("%s", str + 2); return 0; }</td></tr><tr><td>5.</td><td></td><td>CO1 CO3</td></tr></table>			1. #include <stdio.h> void main() { int i, m, n, s=0; int mat[3][3]={1,2,3,4,5,6,7,8,9}; for(i=0;i<3;i++) s=s+mat[i][i]; printf("The sum is %d", s); }	2. #include <stdio.h> int main() { int a[] = {10, 20, 30, 40}; int *p = &a[3]; int *q = &a[1]; printf("%ld", p - q); return 0; }	3. #include <stdio.h> int main() { int a[] = {1, 2, 3}; int *p = a; printf("%d %d", ++*p, *p); return 0; }	4. #include <stdio.h> int main() { char str[] = "abcdef"; printf("%s", str + 2); return 0; }	5.		CO1 CO3							
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	<pre>#include <stdio.h> void modify(char *p) { p = "Programming"; printf("%s ",p); } int main() { char str[] = "Hello"; modify(str); printf("%s", str); return 0; }</pre>		
Que(3)		(10 marks)	
(a)	Explain at least five string manipulation library functions supported in C along with their syntax. Also, write a C program to read a string from the user and then reverse it without using an additional string and display the modified string.		
	Sample Input	Inputted String: Programming is easy	
	Sample Output	ysae si gnimmargorP	
OR			
(b)	Arun stores name of n employees in a single string. His boss calls him and asks him to give a count of all the names starting with the alphabet given by him. Please help Arun to write a code so that he can easily find the count.		
	Sample Input	Inputted String: Ajay Manan Rajat Tejus Aarya Akshit Rashi Anuj	
		Inputted Alphabet: R	
	Sample Output	Total words: 2	
Que(4)		(10 marks)	
(a)	Draw a flowchart to pass three test marks of a student into a function and function will return sum of test marks, average test marks and grade of student. Grade criteria: if average >=75 then Grade = A if average >=50 then Grade = B Otherwise Grade = C		
OR			
(b)	Draw a flowchart to pass a string <i>str</i> into a function named <i>encrypt()</i> . The function will encrypt the string according to following criteria. Display the final encrypted string in <i>main()</i> function. 1. All the vowels will replace by their capital case. 2. All the consonant will replace by their predecessor characters. 3. All the digits will be replaced by '#' character.		
	Sample Input	Sample Output	
	Inputted String: My house number is 157	Output String: Lx gOUrE mUlaEq Ir ###	
Que(5)		(10 marks)	
(a)	Explain the following terms with a snippet of C code wherever possible. i. Pointer to an array ii. NULL pointer iii. Wild Pointer iv. Dangling Pointer v. Memory Leak		
OR			
(b)	Develop a code in C which passes a 1D array of <i>n</i> integer elements into a function <i>replace()</i> as a pointer. Function will replace all elements of array with their factorial. Print the final updated array in <i>main()</i> function.		